

L7c: Ordinary Least Squares

Linear models, the SVD solution, and parameter uncertainty
CHEME 5800 Cornell University

Objectives for Today

How do we estimate the parameters of a linear model from data, and how certain are those estimates?

Today's concepts

- **Estimate linear model parameters with ordinary least squares:** derive the estimate for more observations than parameters, and the least-norm estimate for fewer.
- **Compute the estimate with the singular value decomposition:** write it as a weighted sum of right singular vectors and compare it with solving the normal equations.
- **Quantify the uncertainty in the estimated parameters:** estimate the error variance, compute standard errors, and build confidence intervals.

Lecture: Ordinary Least Squares

Lecture and Lab

Lecture: Ordinary Least Squares

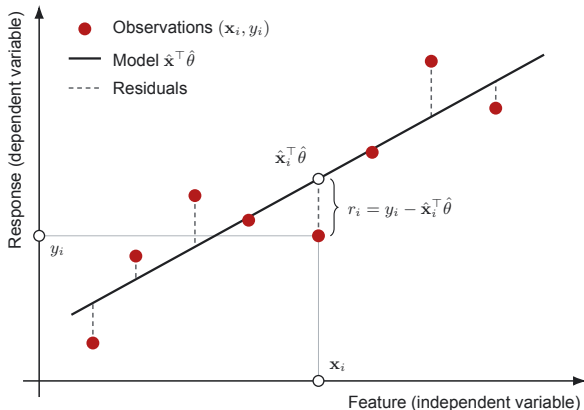
In this lecture, we derive the least-squares estimate of a linear model's parameters, compute it with the singular value decomposition, and quantify its uncertainty.

Lab: L7d

In the lab, we build a linear model of housing prices and estimate its parameters with ordinary least squares.

A linear model describes a continuous output as a weighted sum of input features plus a random error.

Observations, Model, and Residuals



Each dashed segment is a **residual**. Least squares makes the sum of the squared residuals as small as possible.

A Linear Model

Suppose a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i) \mid i = 1, \dots, n\}$ has n labeled examples, with features $\mathbf{x}_i \in \mathbb{R}^m$ and a response $y_i \in \mathbb{R}$. A **linear model** for $\mathcal{D}$ is:

$$y_i = \hat{\mathbf{x}}_i^\top \theta + \epsilon_i, \quad i = 1, 2, \dots, n,$$

where the augmented features $\hat{\mathbf{x}}_i^\top = (x_{i1}, \dots, x_{im}, 1)$ add a 1 for the intercept, $\theta \in \mathbb{R}^p$ with $p = m + 1$ holds the unknown parameters, and ϵ_i is the unobserved random error. In matrix-vector form:

$$\mathbf{y} = \hat{\mathbf{X}} \theta + \epsilon,$$

where the rows of $\hat{\mathbf{X}} \in \mathbb{R}^{n \times p}$ are the $\hat{\mathbf{x}}_i^\top$, and $\mathbf{y}$ and ϵ stack the y_i and ϵ_i .

A linear model only needs to be linear in the **parameters**: polynomial features such as $1, x, x^2, x^3, \dots$ still give a linear model.

Overdetermined Case: The Normal Equations

Suppose $\hat{\mathbf{X}}$ is **overdetermined**, $n > p$, and the errors are $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ with variance σ^2 . We minimize the sum of squared errors:

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{2} \|\mathbf{y} - \hat{\mathbf{X}}\theta\|_2^2.$$

Setting the gradient to zero gives the **normal equations**:

$$\hat{\mathbf{X}}^\top (\mathbf{y} - \hat{\mathbf{X}}\hat{\theta}) = \mathbf{0}.$$

The residual is orthogonal to every column of $\hat{\mathbf{X}}$, so the fitted values $\hat{\mathbf{X}}\hat{\theta}$ are the **orthogonal projection** of $\mathbf{y}$ onto the column space of $\hat{\mathbf{X}}$. With full column rank, $r(\hat{\mathbf{X}}) = p$:

$$\hat{\theta} = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \mathbf{y}.$$

The OLS Estimate Is Unbiased

Substituting the model $\mathbf{y} = \hat{\mathbf{X}}\theta + \epsilon$ into the solution:

$$\hat{\theta} = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top (\hat{\mathbf{X}}\theta + \epsilon) = \theta + (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \epsilon.$$

- **Unbiased estimator**: the errors have mean zero, so $\mathbb{E}[\hat{\theta}] = \theta$. On average, we get the right answer.
- **Parameter uncertainty**: because ϵ is random, $\hat{\theta}$ is a random vector with its own distribution. We quantify it with standard errors and confidence intervals.
- **Bayesian connection**: Bayesian linear regression goes one step further and treats the parameters themselves as random variables.

Underdetermined Case: Least-Norm Solution

Suppose $\hat{\mathbf{X}}$ is **underdetermined**, $n < p$, with full row rank, $r(\hat{\mathbf{X}}) = n$. Infinitely many parameter vectors satisfy $\hat{\mathbf{X}}\theta = \mathbf{y}$ exactly, and we choose the smallest:

$$\min_{\theta} \|\theta\|_2^2 \quad \text{subject to} \quad \hat{\mathbf{X}}\theta = \mathbf{y}.$$

Because $\hat{\mathbf{X}}\hat{\mathbf{X}}^\top$ is invertible, the least-norm solution is:

$$\hat{\theta} = \hat{\mathbf{X}}^\top (\hat{\mathbf{X}}\hat{\mathbf{X}}^\top)^{-1} \mathbf{y} = \mathbf{P} \theta + \hat{\mathbf{X}}^\top (\hat{\mathbf{X}}\hat{\mathbf{X}}^\top)^{-1} \epsilon, \quad \mathbf{P} = \hat{\mathbf{X}}^\top (\hat{\mathbf{X}}\hat{\mathbf{X}}^\top)^{-1} \hat{\mathbf{X}}.$$

The matrix $\mathbf{P}$ is the orthogonal projection onto the row space of $\hat{\mathbf{X}}$. Since $\mathbb{E}[\hat{\theta}] = \mathbf{P}\theta$, the estimate is **biased** unless θ already lies in that subspace.

Why Choose the Smallest Solution?

Every exact solution is the least-norm solution $\hat{\theta}$ plus a vector $\mathbf{z}$ in the **null space** of $\hat{\mathbf{X}}$, so $\hat{\mathbf{X}}\mathbf{z} = \mathbf{0}$. The least-norm solution lies in the row space, which is orthogonal to the null space:

$$\|\hat{\theta} + \mathbf{z}\|_2^2 = \|\hat{\theta}\|_2^2 + \|\mathbf{z}\|_2^2.$$

Adding $\mathbf{z}$ makes the parameter vector longer without changing the predictions at the observed data. The least-norm solution keeps only the part of θ that the data determine.

Regularization adds a penalty to the loss function to prevent overfitting; Ridge (L2) and Lasso (L1) are common choices. The least-norm solution has the same preference for small parameters as L2 regularization.

SVD Solution of the Least-Squares Problem

The SVD of the data matrix is $\hat{\mathbf{X}} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$, with orthogonal $\mathbf{U} \in \mathbb{R}^{n \times n}$ and $\mathbf{V} \in \mathbb{R}^{p \times p}$, and the singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$, $r = r(\hat{\mathbf{X}})$, on the diagonal of $\mathbf{S} \in \mathbb{R}^{n \times p}$. The least-squares estimate is:

$$\hat{\theta} = \mathbf{V}\mathbf{S}^\dagger\mathbf{U}^\top\mathbf{y} \quad \Rightarrow \quad \boxed{\hat{\theta} = \sum_{i=1}^{r(\hat{\mathbf{X}})} \left(\frac{\mathbf{u}_i^\top \mathbf{y}}{\sigma_i} \right) \mathbf{v}_i}$$

where the Moore–Penrose pseudoinverse $\mathbf{S}^\dagger \in \mathbb{R}^{p \times n}$ transposes $\mathbf{S}$ and replaces each nonzero σ_i with $1/\sigma_i$, and $\mathbf{u}_i$ and $\mathbf{v}_i$ are the columns of $\mathbf{U}$ and $\mathbf{V}$.

The same formula gives the OLS estimate under full column rank and the least-norm estimate under full row rank.

Reading the SVD Solution

The estimate is a weighted sum of the right singular vectors $\mathbf{v}_i$. Each weight is the dot product $\mathbf{u}_i^\top \mathbf{y}$ divided by the singular value σ_i .

- **Mode contribution:** each term projects $\mathbf{y}$ onto the data-space direction $\mathbf{u}_i$ and maps the result onto the parameter-space direction $\mathbf{v}_i$.
- **Noise amplification:** when σ_i is small, the factor $1/\sigma_i$ magnifies measurement errors in $\mathbf{y}$ along $\mathbf{v}_i$. This identifies unreliable components of the solution and motivates regularization that dampens the contributions of small singular values.

SVD versus Direct Methods

- **Rank-deficient matrices:** without full column rank, $\hat{\mathbf{X}}^\top \hat{\mathbf{X}}$ is singular. The SVD skips the zero singular values and returns the minimum-norm least-squares solution.
- **Ill-conditioned problems:** the condition number κ is the ratio of the largest to the smallest singular value. With full column rank, $\kappa(\hat{\mathbf{X}}^\top \hat{\mathbf{X}}) = \kappa(\hat{\mathbf{X}})^2$, so forming $\hat{\mathbf{X}}^\top \hat{\mathbf{X}}$ squares it; the SVD factors $\hat{\mathbf{X}}$ directly.
- **Cost:** the thin SVD costs $O(\min(np^2, n^2p))$; the direct method costs $O(np^2)$ to form $\hat{\mathbf{X}}^\top \hat{\mathbf{X}}$ and $O(p^3)$ to solve. For $n \gg p$, both scale as $O(np^2)$, and the direct method is often faster.

For a rectangular matrix $\mathbf{X}$, Julia's backslash operator, $\mathbf{X} \setminus \mathbf{y}$, solves least squares from a pivoted QR factorization of $\mathbf{X}$ and never forms $\mathbf{X}^\top \mathbf{X}$.

The Error Model

We assume $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$: each error is independent and normal, with mean zero and constant variance σ^2 .

- **Normality** gives the estimates exact distributions, for confidence intervals and hypothesis tests. With zero-mean, uncorrelated, constant-variance errors, OLS is the best linear unbiased estimator (BLUE).
- **Uncorrelated errors**: $\sigma^2 \mathbf{I}$ is diagonal, so $\text{Cov}(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$. The parameter variance relies on this structure.
- **Reality check**: by the Central Limit Theorem, the estimates are often approximately normal for large samples, even when the errors are not.

From here on, $\hat{\mathbf{X}}$ is overdetermined with full column rank, and we treat it as fixed, so all of the randomness in $\hat{\theta}$ comes from ϵ .

Estimating the Error Variance

Since the true variance σ^2 is unknown, we estimate it from the residuals $\mathbf{r} = \mathbf{y} - \hat{\mathbf{X}}\hat{\boldsymbol{\theta}}$:

$$\hat{\sigma}^2 = \frac{\|\mathbf{r}\|_2^2}{n-p} = \frac{1}{n-p} \sum_{i=1}^n r_i^2, \quad r_i = y_i - \hat{\mathbf{x}}_i^\top \hat{\boldsymbol{\theta}}.$$

Dividing by $n - p$ accounts for the degrees of freedom used up by estimating p parameters, which makes $\hat{\sigma}^2$ an **unbiased** estimator of σ^2 .

This is why we need $n > p$: with full row rank and $n \leq p$, the fit passes through every observation, so the residuals are zero and carry no information about σ^2 .

Parameter Uncertainty

Since $\hat{\theta} = \theta + (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \epsilon$ and θ is constant, we use $\text{Var}(\mathbf{A}\mathbf{z}) = \mathbf{A} \text{Var}(\mathbf{z}) \mathbf{A}^\top$ for a constant matrix $\mathbf{A}$ and a random vector $\mathbf{z}$, with $\text{Var}(\epsilon) = \sigma^2 \mathbf{I}$:

$$\text{Var}(\hat{\theta}) = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top (\sigma^2 \mathbf{I}) \hat{\mathbf{X}} (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} = \sigma^2 (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1}.$$

The **standard error** of $\hat{\theta}_j$ is its estimated standard deviation, with the unknown σ^2 replaced by $\hat{\sigma}^2$:

$$\text{SE}(\hat{\theta}_j) = \sqrt{\hat{\sigma}^2 [(\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1}]_{jj}}.$$

What Standard Errors Give Us

- **Confidence intervals:** for large samples, $\hat{\theta}_j \pm 1.96 \text{SE}(\hat{\theta}_j)$ is an approximate 95% confidence interval for θ_j . For finite samples, we use the t-distribution.
- **Hypothesis tests:** to test whether $\theta_j = 0$, compare $t_j = \hat{\theta}_j / \text{SE}(\hat{\theta}_j)$ with a t-distribution with $n - p$ degrees of freedom. The p-value is the probability of a value at least this extreme if θ_j is actually 0.
- **Prediction intervals:** the margin around a new prediction combines the parameter covariance $\hat{\sigma}^2(\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1}$ with the error variance $\hat{\sigma}^2$ of the new observation.

Confidence Intervals

A 95% confidence interval contains the true θ_j in about 95% of repeated experiments. Under the normal, constant-variance error model:

$$T_j = \frac{\hat{\theta}_j - \theta_j}{\text{SE}(\hat{\theta}_j)} \sim t_\nu, \quad \nu = n - p.$$

Let $c = t_{1-\alpha/2, \nu}$, the $(1 - \alpha/2)$ quantile of the t-distribution, where α is the significance level ($\alpha = 0.05$ for 95%). Inverting $\Pr(-c \leq T_j \leq c) = 1 - \alpha$ bounds θ_j :

$$\boxed{\hat{\theta}_j \pm t_{1-\alpha/2, \nu} \hat{\sigma} \sqrt{[(\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1}]_{jj}}}$$

One-sided intervals follow the same way from one tail.

Summary

We estimated the parameters of a linear model with least squares, computed the estimate with the SVD, and quantified its uncertainty.

Key takeaways

- **Overdetermined and underdetermined problems.** OLS projects the data onto the column space; with too few observations, we take the least-norm solution.
- **Least squares with the SVD.** The estimate is a weighted sum of right singular vectors, and small singular values amplify noise.
- **Parameter uncertainty.** The residuals give the error variance, which gives standard errors and confidence intervals.

In the L7d lab, we fit a linear model of housing prices.